

MAT 535 HW01

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Problem 1 (Appendix II §1 #2). Let H be a group. Define a map $\mathcal{H} \times$ from **Grp** to itself on objects and morphisms as follows:

$$\mathcal{H} \times (G) = H \times G, \quad \mathcal{H} \times (\varphi: G_1 \rightarrow G_2) = \{(h, g) \mapsto (h, \varphi(g))\}.$$

Prove that $\mathcal{H} \times$ is a functor.

Solution. We only need to check that for any group G , $\mathcal{H} \times (\text{id}_G) = \text{id}_{H \times G}$ and, for homomorphisms $\varphi: G_1 \rightarrow G_2$ and $\psi: G_2 \rightarrow G_3$, $\mathcal{H} \times (\psi \circ \varphi) = \mathcal{H} \times (\psi) \circ \mathcal{H} \times (\varphi)$. The former is true because $\mathcal{H} \times (\text{id}_G)$ is the map sending $(h, g) \in H \times G$ to $(h, \text{id}_G(g)) = (h, g)$, and the latter is true because for any $(h, g) \in H \times G_1$,

$$\mathcal{H} \times (\psi) \circ \mathcal{H} \times (\varphi)(h, g) = \mathcal{H} \times (\psi)(h, \varphi(g)) = (h, \psi \circ \varphi(g)) = \mathcal{H} \times (\psi \circ \varphi)(h, g). \quad \square$$

Problem 2 (Appendix II §1 #3). Show that the map **Ring** to **Grp** by mapping a ring to its group of units defines a functor. Show by explicit examples that this functor is neither faithful nor full.

Solution. Consider the map from **Ring** to **Grp** sending each ring R to its group of units $R^\times$ and sending the ring homomorphism $\varphi: R_1 \rightarrow R_2$ to the group homomorphism $\varphi|_{R_1^\times}: R_1^\times \rightarrow R_2^\times$, which is well-defined because if $r \in R_1$ is a unit then $\varphi(r)\varphi(r^{-1}) = 1_{R_2}$ implies that $\varphi(r)$ is also a unit. This is a functor because the identity map on a ring R of course restricts to the identity on $R^\times$, and given two ring homomorphisms $\varphi: R_1 \rightarrow R_2$ and $\psi: R_2 \rightarrow R_3$,

$$(\psi \circ \varphi)|_{R_1^\times} = \psi \circ (\varphi|_{R_1^\times}) = \psi|_{R_2^\times} \circ \varphi|_{R_1^\times}$$

where we are using the fact that $\varphi|_{R_1^\times} \subseteq R_2^\times$ to get the second equality.

Denote the functor above by $\mathcal{F}$. To show that this functor is neither full nor faithful, consider the map

$$\text{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x], \mathbb{Z}) \xrightarrow{\mathcal{F}} \text{Hom}_{\mathbf{Grp}}(\{1, -1\}, \{1, -1\}).$$

This map sends every homomorphism $\varphi: \mathbb{Z}[x] \rightarrow \mathbb{Z}$ to the identity on $\{1, -1\}$ because $\varphi(1) = 1$ implies $\varphi(-1) = -1$; since there is more than one homomorphism $\varphi: \mathbb{Z}[x] \rightarrow \mathbb{Z}$ (there is a bijection between integers n and homomorphisms φ with $\varphi(x) = n$), we find that $\mathcal{F}$ is not faithful. Of course, the map sending both 1 and -1 to 1 is also a group homomorphism, so $\mathcal{F}$ is not full, either. $\square$

Problem 3 (Appendix II §2 #2). Let H and K be groups and let $\mathcal{H}\times$ and $\mathcal{K}\times$ be the functors on **Grp** defined in the previous section. Let $\varphi: H \rightarrow K$ be a group homomorphism.

- (a) Show that the maps $\eta_A: H \times A \rightarrow K \times A$ given by $\eta_A(h, a) = (\varphi(h), a)$ determine a natural transformation η from $\mathcal{H}\times$ to $\mathcal{K}\times$.
- (b) Show that the transformation η is a natural isomorphism if and only if φ is a group isomorphism.

Solution. (a) To show that η is a natural transformation, we need to show that

$$\eta_B \circ \mathcal{H}\times(\psi) = \mathcal{K}\times(\psi) \circ \eta_A$$

for all group homomorphisms $\psi: A \rightarrow B$. This follows from

$$\eta_B(\{\mathcal{H}\times(\psi)\}(h, a)) = \eta_B(h, \psi(a)) = (\varphi(h), \psi(a)) = \{\mathcal{K}\times(\psi)\}(\varphi(h), a) = \{\mathcal{K}\times(\psi)\}(\eta_A(h, a))$$

for all $(h, a) \in H \times A = \mathcal{H}\times(A)$.

(b) If η is a natural isomorphism for all groups A , then, taking A to be the trivial group, $\eta_1: H \times 1 \rightarrow K \times 1$ is an isomorphism. Since the projections $p_H: H \times 1 \rightarrow H$ and $p_K: K \times 1 \rightarrow K$ onto the first coordinate are isomorphisms, we get that

$$\varphi = p_K \circ \eta_1 \circ p_H^{-1}: H \rightarrow K$$

is an isomorphism, where the equality comes from

$$p_K \circ \eta_1 \circ p_H^{-1}(h) = p_K(\eta_1(h, 1)) = p_K(\varphi(h), 1) = \varphi(h).$$

Conversely, if φ is an isomorphism, then $\eta_A(h, a) = (\varphi(h), a)$ has an inverse $\eta_A^{-1}(k, a) = (\varphi^{-1}(k), a)$, so each η_A is an isomorphism, meaning η is a natural isomorphism of $\mathcal{H}\times$ and $\mathcal{K}\times$. $\square$

Problem 4 (Appendix II §2 #3). Express the universal property of the commutator quotient group as a universal arrow for some functor $\mathcal{F}$.

Solution. Let $\mathcal{F}$ be the functor from **Ab** to **Grp** sending each abelian group and homomorphism of abelian groups to itself. For a group G , let $U(G) = G/G'$ and $\iota: G \rightarrow \mathcal{F}U(G) = U(G)$ be the projection homomorphism. Then, for any abelian group H and group homomorphism $\varphi: G \rightarrow \mathcal{F}H = H$, the universal property of the commutator quotient group dictates that there is a unique morphism of abelian groups $\Phi: G/G' \rightarrow H$ such that $\mathcal{F}(\Phi)\iota = \varphi$, so $(U(G), \iota)$ is a universal arrow from X to the “inclusion functor” $\mathcal{F}$. $\square$